

CHAPTER 8

Mechanical Properties of Solids

Stress, Strain and Young's Modulus

- The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8 \text{ N m}^{-2}$ and $2 \times 10^{11} \text{ N m}^{-2}$, is: (2024)
a. 40 mm b. 8 mm c. 4 mm d. 0.4 mm
- Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is: (2023)
a. Zero b. $\frac{2W}{A}$ c. $\frac{W}{A}$ d. $\frac{W}{2A}$
- The physical quantity that has the same dimensional formula as pressure is: (2022)
a. Coefficient of viscosity
b. Force
c. Momentum
d. Young's modulus of elasticity
- A wire of length L area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is: (2020)
a. $\frac{Mg(L_1 - L)}{AL}$ b. $\frac{MgL}{AL_1}$
c. $\frac{MgL}{A(L_1 - L)}$ d. $\frac{MgL_1}{AL}$
- Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area $3A$. If the length of the first wire is increased by Δl on applying a force F , how much force is needed to stretch the second wire by the same amount? (2018)
a. $4F$ b. $6F$ c. $9F$ d. F

- The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of: (2015 Re)
a. 1 : 1 b. 1 : 2 c. 2 : 1 d. 4 : 1
- Copper of fixed volume V is drawn into wire of length ℓ . When this wire is subjected to a constant force F , the extension produced in the wire is D_1 . Which of the following graphs is a straight line? (2014)
a. $\Delta \ell$ versus $1/\ell$ b. $\Delta \ell$ versus ℓ^2
c. $\Delta \ell$ versus $1/\ell^2$ d. $\Delta \ell$ versus ℓ^3
- The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied? (2013)
a. Length = 300 cm, diameter = 3 mm
b. Length = 50 cm, diameter = 0.5 mm
c. Length = 100 cm, diameter = 1 mm
d. Length = 200 cm, diameter = 2 mm

Bulk Modulus and Shear Modulus

- The bulk modulus of a spherical objects is 'B'. If it is subjected to uniform pressure 'P', the fractional decrease in radius is: (2017-Delhi)
a. $\frac{B}{3p}$ b. $\frac{3P}{B}$ c. $\frac{P}{3B}$ d. $\frac{P}{B}$
- The approximate depth of an ocean is 2700 m. The compressibility of water is $45.4 \times 10^{-11} \text{ Pa}^{-1}$ and density of water is 10^3 kg/m^3 . What fractional compression of water will be obtained at the bottom of the ocean? (2015)
a. 1.0×10^{-2} b. 1.2×10^{-2}
c. 1.4×10^{-2} d. 0.8×10^{-2}

Answer Key

1. (c) 2. (c) 3. (d) 4. (c) 5. (c) 6. (c) 7. (b) 8. (b) 9. (c) 10. (b)